

Differential & Integral Calculus

Complete Worked Solutions for Exam Preparation

Prepared for exam revision (based on Md. Monir Hossain's question bank)

How to use this document: Every question from your sheet is answered below in the order it appears, organised by topic. Each answer is written the way you should write it in the exam: definition/rule first, then the step-by-step working, then a clearly boxed final result. Read the short "idea" line before each worked answer – it explains the trick in plain language so you understand *why* the method works, not just what to write.

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1 Function, Domain & Range

Q1(a). Define function with example. Describe the various types of function.

Answer / How to write in exam

Definition: A function f from a set A to a set B , written $f : A \rightarrow B$, is a rule that assigns to every element $x \in A$ exactly one element $y = f(x) \in B$. A is called the domain, B the co-domain, and the set of actual outputs the range.

Example: $f : \mathbb{R} \rightarrow \mathbb{R}$, $f(x) = x + 2$. Every real x gives exactly one output.

Types of function (write 4–5 with one line each):

- **One-one (injective):** different inputs give different outputs.
- **Onto (surjective):** every element of co-domain is hit by some input.
- **Bijective:** both one-one and onto.
- **Many-one:** at least two inputs give the same output.
- **Into:** range is a proper subset of co-domain.
- **Even function:** $f(-x) = f(x)$ (symmetric about y -axis).
- **Odd function:** $f(-x) = -f(x)$ (symmetric about origin).
- **Algebraic function:** built from $+$, $-$, $\times$, $\div$, powers, roots (e.g. $x^2 + 1$).
- **Transcendental function:** exponential, logarithmic, trigonometric functions.
- **Composite function:** $f \circ g(x) = f(g(x))$.

Q1(b). Test whether the following functions are even or odd.

Answer / How to write in exam

Idea: compute $f(-x)$. If $f(-x) = f(x)$ it's even; if $f(-x) = -f(x)$ it's odd.

i. $f(x) = 1 - \frac{1}{x^2}$

$$f(-x) = 1 - \frac{1}{(-x)^2} = 1 - \frac{1}{x^2} = f(x)$$

$\Rightarrow f(-x) = f(x)$, so f is **even**.

ii. $f(x) = x + \frac{1}{x}$

$$f(-x) = -x + \frac{1}{-x} = -\left(x + \frac{1}{x}\right) = -f(x)$$

$\Rightarrow f$ is **odd**.

iii. $f(x) = \ln(x + \sqrt{1 + x^2})$

$$f(-x) = \ln(-x + \sqrt{1 + x^2})$$

Rationalise: $(\sqrt{1 + x^2} - x)(\sqrt{1 + x^2} + x) = 1 + x^2 - x^2 = 1$, so

$$\sqrt{1 + x^2} - x = \frac{1}{x + \sqrt{1 + x^2}}$$

$$\Rightarrow f(-x) = \ln \frac{1}{x + \sqrt{1 + x^2}} = -\ln(x + \sqrt{1 + x^2}) = -f(x)$$

$\Rightarrow f$ is **odd**.

Q2(a). Define domain & range of a function with example.

Answer / How to write in exam

Domain: the complete set of x -values for which $f(x)$ is defined (well-defined, real, finite).

Range: the complete set of output values $f(x)$ actually takes as x runs over the domain.

Example: $f(x) = \sqrt{x}$. Domain = $[0, \infty)$ (can't take square root of negatives); Range = $[0, \infty)$.

Q2(b). Find the domain & range of the following functions.

Answer / How to write in exam

General rules to remember: (1) denominator $\neq 0$, (2) expression under an even root ≥ 0 , (3) argument of log > 0 .

i. $f(x) = x + 3$ — polynomial, defined everywhere.

Domain = $\mathbb{R}$, Range = $\mathbb{R}$.

ii. $f(x) = x^2 + x + 2$ — complete the square: $x^2 + x + 2 = (x + \frac{1}{2})^2 + \frac{7}{4}$, minimum value $\frac{7}{4}$ at $x = -\frac{1}{2}$.

Domain = $\mathbb{R}$, Range = $[\frac{7}{4}, \infty)$.

iii. $f(x) = \sqrt{x+3}$ — need $x+3 \geq 0$.

Domain = $[-3, \infty)$, Range = $[0, \infty)$.

iv. $f(x) = \sqrt{16 - x^2}$ — need $16 - x^2 \geq 0 \Rightarrow -4 \leq x \leq 4$.

Domain = $[-4, 4]$, Range = $[0, 4]$.

v. $f(x) = \sqrt{x^2 - 9}$ — need $x^2 \geq 9 \Rightarrow x \leq -3$ or $x \geq 3$.

Domain = $(-\infty, -3] \cup [3, \infty)$, Range = $[0, \infty)$.

vi. $f(x) = \sqrt{\frac{x+2}{x-2}}$ — need $\frac{x+2}{x-2} \geq 0$ and $x \neq 2$. Critical points $-2, 2$; sign analysis gives $x \leq -2$ or $x > 2$.

Domain = $(-\infty, -2] \cup (2, \infty)$.

For range put $y = \sqrt{\frac{x+2}{x-2}}$ ($y \geq 0$): $y^2(x-2) = x+2 \Rightarrow x = \frac{2(y^2+1)}{y^2-1}$, undefined at $y = 1$.

Range = $[0, 1) \cup (1, \infty)$.

vii. $f(x) = \sqrt{\frac{1-x}{x}}$ — need $\frac{1-x}{x} \geq 0$, $x \neq 0$. Sign chart $\Rightarrow 0 < x \leq 1$.

Domain = $(0, 1]$, Range = $[0, \infty)$ (as $x \rightarrow 0^+$, $f \rightarrow \infty$; at $x = 1$, $f = 0$).

viii. $f(x) = \frac{x+1}{x-1}$ — Domain: $x \neq 1$, so $\mathbb{R} \setminus \{1\}$.

Range: solve $y = \frac{x+1}{x-1} \Rightarrow x = \frac{y+1}{y-1}$, undefined at $y = 1$. Range = $\mathbb{R} \setminus \{1\}$.

ix. $f(x) = \frac{x-3}{2x+1}$ — Domain: $x \neq -\frac{1}{2}$.

Range: $x = \frac{-3-y}{2y-1}$, undefined at $y = \frac{1}{2}$. Range = $\mathbb{R} \setminus \{\frac{1}{2}\}$.

x. $f(x) = \frac{x}{|x|}$ — equals $+1$ for $x > 0$, -1 for $x < 0$.

Domain = $\mathbb{R} \setminus \{0\}$, Range = $\{-1, 1\}$.

xi. $f(x) = |x| - x$ — for $x \geq 0$: 0 ; for $x < 0$: $-2x > 0$.

Domain = $\mathbb{R}$, Range = $[0, \infty)$.

xii. $f(x) = |x+1| + |x-1|$ — for $x \leq -1$: $-2x$; for $-1 \leq x \leq 1$: constant 2 ; for $x \geq 1$: $2x$.

Domain = $\mathbb{R}$, Range = $[2, \infty)$.

xiii. $f(x) = |x| + |x - 1|$ — minimum value = 1 (distance between 0 and 1), attained on $[0, 1]$.

Domain = $\mathbb{R}$, Range = $[1, \infty)$.

xiv. $f(x) = |x + 2| + |x - 2|$ — minimum = 4 (distance between $-2, 2$).

Domain = $\mathbb{R}$, Range = $[4, \infty)$.

xv. $f(x) = |x + 3| + |x - 3|$ — minimum = 6.

Domain = $\mathbb{R}$, Range = $[6, \infty)$.

xvi. $f(x) = |x + 1| + |x| + |x - 1|$ — sum of distances from $-1, 0, 1$; minimised at the middle point $x = 0$ giving value $1 + 0 + 1 = 2$.

Domain = $\mathbb{R}$, Range = $[2, \infty)$.

$$\textbf{xvii. } f(x) = \begin{cases} 3, & -3 \leq x < -1 \\ -6x - 3, & -1 \leq x \leq 0 \\ 3x - 3, & 0 < x \leq 1 \end{cases}$$

Domain (given) = $[-3, 1]$. On $[-3, -1]$: value 3. On $[-1, 0]$: linear from $f(-1) = 3$ down to $f(0) = -3$, covering $[-3, 3]$. On $(0, 1]$: linear from just above -3 up to $f(1) = 0$.

Range = $[-3, 3]$.

Q3. Draw the graph of the following functions (with domain, range and key features to plot).

Answer / How to write in exam

Exam tip: you are not expected to plot hundreds of points – find domain, range, intercepts, symmetry and 3–4 key points, then join smoothly. Key data for each:

- i.** $f(x) = \sqrt{x}$: Domain $[0, \infty)$, Range $[0, \infty)$. Passes through $(0, 0), (1, 1), (4, 2)$. Increasing, concave down curve starting at origin.
- ii.** $f(x) = \frac{1}{x}$: Domain/Range $\mathbb{R} \setminus \{0\}$. Rectangular hyperbola in 1st & 3rd quadrants; asymptotes $x = 0, y = 0$.
- iii.** $f(x) = -x^2 + 1$: Downward parabola, vertex $(0, 1)$, roots at $x = \pm 1$.
- iv.** $f(x) = 3 - 2x - x^2 = -[(x + 1)^2 - 4]$, vertex $(-1, 4)$, downward parabola, roots $x = 1, -3$.
- v.** $f(x) = \begin{cases} x^2 + 1, & x < 0 \\ x, & 0 \leq x \leq 1 \\ 1/x, & x > 1 \end{cases}$: left branch is part of upward parabola shifted up 1 (approaching $(0, 1)$ but jumps to line); middle branch straight line through origin to $(1, 1)$; right branch is hyperbola decreasing from $(1, 1)$ towards $y = 0$. **Note:** function jumps at $x = 0$ (1 vs 0) – mark open/closed circles.
- vi.** $f(x) = \sin x, \cos x, \tan x$: standard periodic curves. $\sin x, \cos x$: period 2π , range $[-1, 1]$. $\tan x$: period π , range $\mathbb{R}$, vertical asymptotes at $x = \pm\pi/2, \pm3\pi/2, \dots$
- vii.** $f(x) = a^x, a > 1$: passes through $(0, 1)$, always positive, increasing, horizontal asymptote $y = 0$ as $x \rightarrow -\infty$.
- viii.** $f(x) = |x| + |x - 1|$: constant = 1 on $[0, 1]$, slope -2 for $x < 0$, slope 2 for $x > 1$ (a "bathtub"/trapezoid shape).
- ix.** $f(x) = \begin{cases} 3 + 2x, & -\frac{3}{2} < x < 0 \\ 3 - 2x, & 0 \leq x < \frac{3}{2} \\ -3 - 2x, & x \geq \frac{3}{2} \end{cases}$: three straight-line segments; joins continuously at $x = 0$

(value 3, a "tent" peak) then at $x = 3/2$ jumps from 0 down to -6 (discontinuity – mark with open/filled circle).

2 Limit

Q1. Define limit of a function.

Answer / How to write in exam

We say $\lim_{x \rightarrow a} f(x) = L$ if the value of $f(x)$ gets arbitrarily close to L as x gets arbitrarily close to a (from either side), without x actually equalling a .

Formal (ε - δ) definition: for every $\varepsilon > 0$ there exists a $\delta > 0$ such that

$$0 < |x - a| < \delta \Rightarrow |f(x) - L| < \varepsilon.$$

The limit exists at $x = a$ only if the **Left Hand Limit (LHL) = Right Hand Limit (RHL)**.

$$\text{Q2(a). } f(x) = \begin{cases} 1 + 2x, & -\frac{1}{2} \leq x < 0 \\ 1 - 2x, & 0 \leq x < \frac{1}{2} \\ 2x - 1, & x > \frac{1}{2} \end{cases} \text{ . Does } \lim_{x \rightarrow 0} f(x) \text{ and } \lim_{x \rightarrow \frac{1}{2}} f(x) \text{ exist?}$$

Answer / How to write in exam

At $x = 0$:

$$\text{LHL} = \lim_{x \rightarrow 0^-} (1 + 2x) = 1, \quad \text{RHL} = \lim_{x \rightarrow 0^+} (1 - 2x) = 1$$

$\text{LHL} = \text{RHL} = 1 \Rightarrow$ **limit exists**, $\lim_{x \rightarrow 0} f(x) = 1$.

At $x = \frac{1}{2}$:

$$\text{LHL} = \lim_{x \rightarrow \frac{1}{2}^-} (1 - 2x) = 1 - 1 = 0, \quad \text{RHL} = \lim_{x \rightarrow \frac{1}{2}^+} (2x - 1) = 1 - 1 = 0$$

$\text{LHL} = \text{RHL} = 0 \Rightarrow$ **limit exists**, $\lim_{x \rightarrow \frac{1}{2}} f(x) = 0$.

$$\text{Q2(b). } f(x) = \begin{cases} x^2, & x < 1 \\ 3, & x = 1 \\ x^2 + 1, & x > 1 \end{cases} \text{ . Does } \lim_{x \rightarrow 1} f(x) \text{ exist?}$$

Answer / How to write in exam

$$\text{LHL} = \lim_{x \rightarrow 1^-} x^2 = 1, \quad \text{RHL} = \lim_{x \rightarrow 1^+} (x^2 + 1) = 1 + 1 = 2$$

$\text{LHL} \neq \text{RHL} \Rightarrow$ **limit does not exist** at $x = 1$.

Q2(c). $f(x) = \begin{cases} x, & x > 0 \\ 0, & x = 0 \\ -x, & x < 0 \end{cases}$. Does $\lim_{x \rightarrow 0} f(x)$ exist?

Answer / How to write in exam

(This function is simply $f(x) = |x|$.)

$$\text{LHL} = \lim_{x \rightarrow 0^-} (-x) = 0, \quad \text{RHL} = \lim_{x \rightarrow 0^+} x = 0$$

$$\text{LHL} = \text{RHL} = 0 \Rightarrow \text{limit exists, } \lim_{x \rightarrow 0} f(x) = 0.$$

Q2(d). Let $f(x) = \frac{3x + |x|}{7x - 5|x|}$. Does $\lim_{x \rightarrow 0} f(x)$ exist?

Answer / How to write in exam

$$\text{For } x > 0, |x| = x: f(x) = \frac{3x + x}{7x - 5x} = \frac{4x}{2x} = 2 \Rightarrow \text{RHL} = 2.$$

$$\text{For } x < 0, |x| = -x: f(x) = \frac{3x - x}{7x + 5x} = \frac{2x}{12x} = \frac{1}{6} \Rightarrow \text{LHL} = \frac{1}{6}.$$

$$\text{LHL} \neq \text{RHL} \Rightarrow \text{limit does not exist at } x = 0.$$

3 Continuity

Q1. Define continuity of a function.

Answer / How to write in exam

$f(x)$ is continuous at $x = a$ if all three conditions hold:

1. $f(a)$ is defined,
2. $\lim_{x \rightarrow a} f(x)$ exists (i.e. $\text{LHL} = \text{RHL}$),
3. $\lim_{x \rightarrow a} f(x) = f(a)$.

In short: **LHL = RHL = $f(a)$.**

Q2. Test the continuity of $f(x) = \begin{cases} 5, & 0 < x < 1 \\ 10, & 1 \leq x \leq 2 \\ 15, & 2 < x \leq 3 \end{cases}$ at $x = 2$.

Answer / How to write in exam

$$\text{LHL} = \lim_{x \rightarrow 2^-} 10 = 10, \quad \text{RHL} = \lim_{x \rightarrow 2^+} 15 = 15, \quad f(2) = 10$$

$$\text{LHL} \neq \text{RHL} \Rightarrow \text{discontinuous at } x = 2.$$

Q3. Test the continuity of $f(x) = \begin{cases} \frac{1 - \cos x}{x^2}, & x \neq 0 \\ 1, & x = 0 \end{cases}$ **at** $x = 0$.

Answer / How to write in exam

Standard limit used: $\lim_{x \rightarrow 0} \frac{1 - \cos x}{x^2} = \frac{1}{2}$ (from $1 - \cos x = 2 \sin^2(x/2)$, so $\frac{1 - \cos x}{x^2} = \frac{2 \sin^2(x/2)}{x^2} = \frac{1}{2} \left(\frac{\sin(x/2)}{x/2} \right)^2 \rightarrow \frac{1}{2}$).

So $\lim_{x \rightarrow 0} f(x) = \frac{1}{2}$, but $f(0) = 1$.

Since limit $\neq f(0)$, f is **discontinuous at** $x = 0$ (a *removable* discontinuity).

Q4. Test the continuity of $f(x) = \begin{cases} 3 + 2x, & -\frac{3}{2} \leq x < 0 \\ 3 - 2x, & 0 \leq x < \frac{3}{2} \\ -3 - 2x, & x \geq \frac{3}{2} \end{cases}$ **at** $x = 0$ **and** $x = \frac{3}{2}$.

Answer / How to write in exam

At $x = 0$:

$$\text{LHL} = \lim_{x \rightarrow 0^-} (3 + 2x) = 3, \quad \text{RHL} = \lim_{x \rightarrow 0^+} (3 - 2x) = 3, \quad f(0) = 3$$

All equal $\Rightarrow$ **continuous at** $x = 0$.

At $x = \frac{3}{2}$:

$$\text{LHL} = \lim_{x \rightarrow \frac{3}{2}^-} (3 - 2x) = 3 - 3 = 0, \quad f\left(\frac{3}{2}\right) = -3 - 2\left(\frac{3}{2}\right) = -6$$

$\text{LHL} \neq f\left(\frac{3}{2}\right) \Rightarrow$ **discontinuous at** $x = \frac{3}{2}$.

Q5. Test the continuity at $x = 0, 1$ **for** $f(x) = \begin{cases} x^2 + 1, & x < 0 \\ x, & 0 \leq x \leq 1 \\ 1/x, & x > 1 \end{cases}$

Answer / How to write in exam

At $x = 0$:

$$\text{LHL} = \lim_{x \rightarrow 0^-} (x^2 + 1) = 1, \quad \text{RHL} = f(0) = 0$$

$\text{LHL} \neq \text{RHL} \Rightarrow$ **discontinuous at** $x = 0$.

At $x = 1$:

$$\text{LHL} = \lim_{x \rightarrow 1^-} x = 1, \quad \text{RHL} = \lim_{x \rightarrow 1^+} \frac{1}{x} = 1, \quad f(1) = 1$$

All equal $\Rightarrow$ **continuous at** $x = 1$.

Q6. Test continuity at $x = 0, 1, 2$ for $f(x) = \begin{cases} -x^2, & x \leq 0 \\ 5x - 4, & 0 < x \leq 1 \\ 4x^2 - 3x, & 1 < x < 2 \\ 3x + 4, & x \geq 2 \end{cases}$

Answer / How to write in exam

At $x = 0$: LHL = $\lim_{x \rightarrow 0^-} (-x^2) = 0$; RHL = $\lim_{x \rightarrow 0^+} (5x - 4) = -4$. Unequal $\Rightarrow$ **discontinuous**.

At $x = 1$: LHL = $\lim_{x \rightarrow 1^-} (5x - 4) = 1$; RHL = $\lim_{x \rightarrow 1^+} (4x^2 - 3x) = 4 - 3 = 1$; $f(1) = 5 - 4 = 1$. All equal $\Rightarrow$ **continuous**.

At $x = 2$: LHL = $\lim_{x \rightarrow 2^-} (4x^2 - 3x) = 16 - 6 = 10$; RHL = $\lim_{x \rightarrow 2^+} (3x + 4) = 10$; $f(2) = 10$. All equal $\Rightarrow$ **continuous**.

Q7. Test the continuity at $x = 0$ for $f(x) = \begin{cases} |x|/x, & x \neq 0 \\ 1, & x = 0 \end{cases}$

Answer / How to write in exam

$$\text{LHL} = \lim_{x \rightarrow 0^-} \frac{-x}{x} = -1, \quad \text{RHL} = \lim_{x \rightarrow 0^+} \frac{x}{x} = 1$$

LHL $\neq$ RHL $\Rightarrow$ **discontinuous at $x = 0$** .

Q8. Test the continuity at $x = 0$ for $f(x) = \begin{cases} -\cos x, & x \neq 0 \\ \cos x, & x = 0 \end{cases}$

Answer / How to write in exam

$$\lim_{x \rightarrow 0} f(x) = \lim_{x \rightarrow 0} (-\cos x) = -1 \quad (\text{using the } x \neq 0 \text{ branch, since limit ignores the actual point})$$

$f(0) = \cos 0 = 1$. Since $-1 \neq 1$, f is **discontinuous at $x = 0$** .

Q9. Test continuity at $x = 0$ for $f(x) = \begin{cases} \frac{\log_e(1+x)}{\tan^{-1} x}, & x \neq 0 \\ 1, & x = 0 \end{cases}$

Answer / How to write in exam

Standard limits: $\lim_{x \rightarrow 0} \frac{\ln(1+x)}{x} = 1$ and $\lim_{x \rightarrow 0} \frac{\tan^{-1} x}{x} = 1$.

$$\lim_{x \rightarrow 0} f(x) = \lim_{x \rightarrow 0} \frac{\ln(1+x)/x}{\tan^{-1} x/x} = \frac{1}{1} = 1 = f(0)$$

Continuous at $x = 0$.

Q10. Test continuity at $x = 1, 2$ for $f(x) = \begin{cases} \log x, & 0 < x \leq 1 \\ 0, & 1 < x < 2 \\ 1 + x^2, & x \geq 2 \end{cases}$

Answer / How to write in exam

At $x = 1$: LHL = $\lim_{x \rightarrow 1^-} \log x = \log 1 = 0$; RHL = $\lim_{x \rightarrow 1^+} 0 = 0$; $f(1) = \log 1 = 0$. All equal $\Rightarrow$ **continuous**.

At $x = 2$: LHL = $\lim_{x \rightarrow 2^-} 0 = 0$; RHL/ $f(2) = 1 + 4 = 5$. Unequal $\Rightarrow$ **discontinuous**.

Q11. Test the continuity of $f(x) = |x| + |x + 1|$ at $x = 1$.

Answer / How to write in exam

Near $x = 1$ ($x > 0$ and $x > -1$), both $|x| = x$ and $|x + 1| = x + 1$, so $f(x) = x + (x + 1) = 2x + 1$, a polynomial — continuous everywhere, in particular at $x = 1$.

$$\lim_{x \rightarrow 1} f(x) = 2(1) + 1 = 3 = f(1) = |1| + |2| = 3$$

Continuous at $x = 1$. (In fact $|x| + |x + 1|$, being a sum of two continuous functions, is continuous for all x .)

4 Differentiability

Q1. Define differentiability of a function.

Answer / How to write in exam

$f(x)$ is differentiable at $x = a$ if

$$f'(a) = \lim_{h \rightarrow 0} \frac{f(a+h) - f(a)}{h}$$

exists and is finite. Equivalently, the **Left Hand Derivative (LHD)** must equal the **Right Hand Derivative (RHD)** at that point.

Key fact: differentiability $\Rightarrow$ continuity, but continuity does *not* imply differentiability (e.g. $|x|$ is continuous but not differentiable at 0). So always check continuity first.

Q2. Test whether $f(x) = x^2 + 5$ is differentiable at $x = 1$.

Answer / How to write in exam

By first principle:

$$f'(1) = \lim_{h \rightarrow 0} \frac{f(1+h) - f(1)}{h} = \lim_{h \rightarrow 0} \frac{(1+h)^2 + 5 - 6}{h} = \lim_{h \rightarrow 0} \frac{2h + h^2}{h} = \lim_{h \rightarrow 0} (2 + h) = 2$$

The limit exists and is finite $\Rightarrow f$ is **differentiable at $x = 1$, with $f'(1) = 2$** . (Every polynomial

is differentiable everywhere.)

Q3. $f(x) = \begin{cases} 5x + 4, & 0 < x \leq 1 \\ 4x^2 - 3x, & 1 < x < 2 \end{cases}$ (i) Test continuity at $x = 1$; (ii) test differentiability at $x = 1$.

Answer / How to write in exam

(i) **Continuity at $x = 1$:**

$$\text{LHL} = \lim_{x \rightarrow 1^-} (5x + 4) = 9, \quad \text{RHL} = \lim_{x \rightarrow 1^+} (4x^2 - 3x) = 4 - 3 = 1$$

$\text{LHL} \neq \text{RHL} \Rightarrow$ **discontinuous at $x = 1$.**

(ii) **Differentiability at $x = 1$:** Since differentiability requires continuity first, and f is **not continuous** at $x = 1$, f is automatically **not differentiable at $x = 1$.**

Q4. Test the differentiability at $x = 0$ of $f(x) = \begin{cases} 3 + 2x, & -\frac{3}{2} < x < 0 \\ 3 - 2x, & 0 \leq x < \frac{3}{2} \\ -3 - 2x, & x \geq \frac{3}{2} \end{cases}$

Answer / How to write in exam

Continuity at $x = 0$: $\text{LHL} = 3$, $\text{RHL} = 3$, $f(0) = 3$ — continuous (checked as in the continuity section).

$$\text{LHD} = \left. \frac{d}{dx}(3 + 2x) \right|_{x=0} = 2. \quad \text{RHD} = \left. \frac{d}{dx}(3 - 2x) \right|_{x=0} = -2.$$

$\text{LHD} \neq \text{RHD} \Rightarrow$ **not differentiable at $x = 0$** (there is a sharp corner there).

Q5. Test the differentiability at $x = \frac{\pi}{2}$ of $f(x) = \begin{cases} 1, & x < 0 \\ 1 + \sin x, & 0 \leq x < \frac{\pi}{2} \\ 2 + (x - \frac{\pi}{2})^2, & x \geq \frac{\pi}{2} \end{cases}$

Answer / How to write in exam

Continuity at $x = \pi/2$: $\text{LHL} = \lim(1 + \sin x) = 1 + 1 = 2$; $\text{RHL} = f(\pi/2) = 2 + 0 = 2$. Continuous.

$$\text{LHD} = \left. \frac{d}{dx}(1 + \sin x) \right|_{x=\pi/2} = \cos\left(\frac{\pi}{2}\right) = 0. \quad \text{RHD} = \left. \frac{d}{dx} \left[2 + (x - \frac{\pi}{2})^2 \right] \right|_{x=\pi/2} = 2 \left(x - \frac{\pi}{2} \right) \Big|_{x=\pi/2} = 0.$$

$\text{LHD} = \text{RHD} = 0 \Rightarrow$ **differentiable at $x = \pi/2$.**

Q6. Test the differentiability at $x = 2$ of $f(x) = \begin{cases} 3x + 2, & x > 2 \\ 12 - x^2, & x \leq 2 \end{cases}$

Answer / How to write in exam

Continuity at $x = 2$: LHL = $12 - 4 = 8$; RHL = $6 + 2 = 8$; $f(2) = 8$. Continuous.

$$\text{LHD} = \frac{d}{dx}(12 - x^2) \Big|_{x=2} = -2x \Big|_{x=2} = -4. \quad \text{RHD} = \frac{d}{dx}(3x + 2) = 3.$$

LHD $\neq$ RHD $\Rightarrow$ **not differentiable at $x = 2$.**

Q7. Test the differentiability at $x = 2$ of $f(x) = \begin{cases} x, & 0 < x < 1 \\ 2 - x, & 1 \leq x \leq 2 \\ x - \frac{1}{2}x^2, & x > 2 \end{cases}$

Answer / How to write in exam

Continuity at $x = 2$: LHL = $2 - 2 = 0$; RHL = $2 - 2 = 0$; $f(2) = 0$. Continuous.

$$\text{LHD} = \frac{d}{dx}(2 - x) = -1. \quad \text{RHD} = \frac{d}{dx} \left(x - \frac{1}{2}x^2 \right) \Big|_{x=2} = (1 - x) \Big|_{x=2} = -1.$$

LHD = RHD = $-1 \Rightarrow$ **differentiable at $x = 2$.**

5 Indeterminate Form (L'Hospital's Rule)

L'Hospital's Rule: if $\lim_{x \rightarrow a} \frac{f(x)}{g(x)}$ gives $\frac{0}{0}$ or $\frac{\infty}{\infty}$, then $\lim_{x \rightarrow a} \frac{f(x)}{g(x)} = \lim_{x \rightarrow a} \frac{f'(x)}{g'(x)}$ (differentiate top and bottom *separately*), repeating if still indeterminate. For $0 \times \infty$, rewrite as a fraction first; for $\infty^0, 0^0, 1^\infty$, take $\ln$ first.

Q1. $\lim_{x \rightarrow 0} \frac{xe^x - \log(1+x)}{x^2}$

Answer / How to write in exam

At $x = 0$: numerator = 0, denominator = 0 $\Rightarrow \frac{0}{0}$ form. Apply L'Hospital:

$$\lim_{x \rightarrow 0} \frac{e^x + xe^x - \frac{1}{1+x}}{2x}$$

At $x = 0$: numerator = $1 + 0 - 1 = 0$, denominator = 0 — still $\frac{0}{0}$. Differentiate again:

$$\lim_{x \rightarrow 0} \frac{e^x + e^x + xe^x + \frac{1}{(1+x)^2}}{2} = \frac{2e^x + xe^x + \frac{1}{(1+x)^2}}{2} \Big|_{x=0} = \frac{2 + 0 + 1}{2} = \boxed{\frac{3}{2}}$$

Q2. $\lim_{x \rightarrow 0} \frac{e^x - e^{-x} + 2 \sin x - 4x}{x^5}$

Answer / How to write in exam

Faster method: Maclaurin series (repeated L'Hospital 5 times is very long).

$$e^x - e^{-x} = 2x + \frac{2x^3}{6} + \frac{2x^5}{120} + \cdots = 2x + \frac{x^3}{3} + \frac{x^5}{60} + \cdots$$

$$2 \sin x = 2x - \frac{2x^3}{6} + \frac{2x^5}{120} - \cdots = 2x - \frac{x^3}{3} + \frac{x^5}{60} - \cdots$$

Adding: $e^x - e^{-x} + 2 \sin x = 4x + \frac{2x^5}{60} + \cdots = 4x + \frac{x^5}{30} + \cdots$

Subtract $4x$: numerator = $\frac{x^5}{30} + (\text{higher order terms})$.

$$\lim_{x \rightarrow 0} \frac{x^5/30 + \cdots}{x^5} = \boxed{\frac{1}{30}}$$

Q3. $\lim_{\theta \rightarrow \frac{\pi}{4}} \frac{\sqrt{2} - \cos \theta - \sin \theta}{(4\theta - \pi)^2}$

Answer / How to write in exam

At $\theta = \pi/4$: $\cos \theta = \sin \theta = \frac{\sqrt{2}}{2}$, so numerator = $\sqrt{2} - \sqrt{2} = 0$; denominator = $(\pi - \pi)^2 = 0$. $\frac{0}{0}$ form.

Differentiate w.r.t. θ :

$$\lim_{\theta \rightarrow \frac{\pi}{4}} \frac{\sin \theta - \cos \theta}{8(4\theta - \pi)}$$

At $\theta = \pi/4$: numerator = 0, denominator = 0 — apply L'Hospital again:

$$\lim_{\theta \rightarrow \frac{\pi}{4}} \frac{\cos \theta + \sin \theta}{32} = \frac{\frac{\sqrt{2}}{2} + \frac{\sqrt{2}}{2}}{32} = \frac{\sqrt{2}}{32}$$

$$\boxed{\frac{\sqrt{2}}{32}}$$

Q4. $\lim_{x \rightarrow a} \frac{x^n - a^n}{x - a}$

Answer / How to write in exam

$\frac{0}{0}$ form. L'Hospital (differentiate w.r.t. x):

$$\lim_{x \rightarrow a} \frac{nx^{n-1}}{1} = \boxed{na^{n-1}}$$

(This is exactly the first-principle definition of the derivative of x^n .)

Q5. $\lim_{x \rightarrow 3} \frac{x^2 - 9}{x - 3}$

Answer / How to write in exam

Factor: $\frac{x^2 - 9}{x - 3} = \frac{(x - 3)(x + 3)}{x - 3} = x + 3$ (for $x \neq 3$).

$$\lim_{x \rightarrow 3} (x + 3) = \boxed{6}$$

Q6. $\lim_{x \rightarrow 0} \frac{\sqrt{1+x} - 1}{x}$

Answer / How to write in exam

Multiply numerator and denominator by the conjugate $\sqrt{1+x} + 1$:

$$\frac{(\sqrt{1+x} - 1)(\sqrt{1+x} + 1)}{x(\sqrt{1+x} + 1)} = \frac{(1+x) - 1}{x(\sqrt{1+x} + 1)} = \frac{1}{\sqrt{1+x} + 1}$$

$$\lim_{x \rightarrow 0} \frac{1}{\sqrt{1+x} + 1} = \boxed{\frac{1}{2}}$$

Q7. $\lim_{x \rightarrow a} \frac{x^{7/2} - a^{7/2}}{\sqrt{x} - \sqrt{a}}$

Answer / How to write in exam

Put $u = \sqrt{x}$ (so $x^{7/2} = u^7$); as $x \rightarrow a$, $u \rightarrow \sqrt{a}$:

$$\lim_{u \rightarrow \sqrt{a}} \frac{u^7 - (\sqrt{a})^7}{u - \sqrt{a}} = 7(\sqrt{a})^6 \quad (\text{same formula as Q4})$$

$$= 7a^3$$

$$\boxed{7a^3}$$

Q8. $\lim_{x \rightarrow 0} \frac{1 - \cos x}{x^2}$

Answer / How to write in exam

Using $1 - \cos x = 2 \sin^2(x/2)$:

$$\frac{2 \sin^2(x/2)}{x^2} = \frac{1}{2} \left(\frac{\sin(x/2)}{x/2} \right)^2 \rightarrow \frac{1}{2}(1)^2$$

$$\boxed{\frac{1}{2}}$$

Q9. $\lim_{x \rightarrow 0} x^2 \log x^2$

Answer / How to write in exam

$0 \times (-\infty)$ form. Rewrite as a fraction: $x^2 \ln x^2 = \frac{\ln x^2}{1/x^2}$, now $\frac{-\infty}{\infty}$. L'Hospital:

$$\lim_{x \rightarrow 0} \frac{2/x}{-2/x^3} = \lim_{x \rightarrow 0} (-x^2) = 0$$

$$\boxed{0}$$

Q10. $\lim_{x \rightarrow 0} \frac{\tan x - \sin x}{x^3}$

Answer / How to write in exam

Series: $\tan x = x + \frac{x^3}{3} + \dots$, $\sin x = x - \frac{x^3}{6} + \dots$

$$\tan x - \sin x = \frac{x^3}{3} + \frac{x^3}{6} + \dots = \frac{x^3}{2} + \dots$$

$$\lim_{x \rightarrow 0} \frac{x^3/2 + \dots}{x^3} = \boxed{\frac{1}{2}}$$

Q11. Evaluate using L'Hospital's rule: $\lim_{x \rightarrow \infty} 2^x \sin\left(\frac{a}{2^x}\right)$

Answer / How to write in exam

Let $t = \frac{1}{2^x}$; as $x \rightarrow \infty$, $t \rightarrow 0$. The expression becomes

$$\frac{\sin(at)}{t}$$

Using the standard limit $\lim_{t \rightarrow 0} \frac{\sin(at)}{t} = a$:

$$\boxed{a}$$

Q12. Evaluate using L'Hospital's rule: $\lim_{x \rightarrow 0} \left(\frac{1}{x}\right)^{\tan x}$

Answer / How to write in exam

∞^0 form. Let $y = \left(\frac{1}{x}\right)^{\tan x}$. Take $\ln$:

$$\ln y = \tan x \cdot \ln \left(\frac{1}{x}\right) = -\tan x \ln x$$

This is $0 \times (-\infty)$; rewrite as $\frac{-\ln x}{\cot x}$ ($\frac{\infty}{\infty}$ form) and apply L'Hospital:

$$\lim_{x \rightarrow 0} \frac{-1/x}{-\csc^2 x} = \lim_{x \rightarrow 0} \frac{\sin^2 x}{x} = \lim_{x \rightarrow 0} \frac{x^2}{x} = 0$$

So $\ln y \rightarrow 0 \Rightarrow y \rightarrow e^0 = 1$.

1

Q13. Evaluate using L'Hospital's rule: $\lim_{x \rightarrow \infty} (e^x + 1)^{-2/x}$

Answer / How to write in exam

1^∞ -type (base $\rightarrow \infty$, exponent $\rightarrow 0$; treat carefully). Let $y = (e^x + 1)^{-2/x}$.

$$\ln y = -\frac{2}{x} \ln(e^x + 1) = -\frac{2}{x} [x + \ln(1 + e^{-x})] = -2 - \frac{2 \ln(1 + e^{-x})}{x}$$

As $x \rightarrow \infty$, $e^{-x} \rightarrow 0$ so $\ln(1 + e^{-x}) \rightarrow 0$, hence the second term $\rightarrow 0$.

$$\ln y \rightarrow -2 \Rightarrow y \rightarrow e^{-2}$$

e^{-2}

6 Differentiation

Q1. Describe the physical meaning of the derivative of a function.

Answer / How to write in exam

The derivative $f'(x)$ is the **instantaneous rate of change** of f with respect to x . Geometrically, it is the **slope of the tangent line** to the curve $y = f(x)$ at that point. Physically, e.g. if $s(t)$ is displacement, $\frac{ds}{dt}$ is velocity (rate of change of position with time); $\frac{dv}{dt}$ is acceleration.

Q2. The radius of a circle increases at a rate of 3 cm/sec. Find the rate of change of the area when (i) the radius is 5 cm, (ii) the area is 4π cm².

Answer / How to write in exam

Given $\frac{dr}{dt} = 3 \text{ cm/s}$. Area $A = \pi r^2$. Differentiate w.r.t. time t (chain rule):

$$\frac{dA}{dt} = 2\pi r \frac{dr}{dt} = 6\pi r$$

(i) At $r = 5$: $\frac{dA}{dt} = 6\pi(5) = \boxed{30\pi \text{ cm}^2/\text{s}}$

(ii) $A = 4\pi \Rightarrow \pi r^2 = 4\pi \Rightarrow r^2 = 4 \Rightarrow r = 2$.

$$\frac{dA}{dt} = 6\pi(2) = \boxed{12\pi \text{ cm}^2/\text{s}}$$

Q3. By definition (first principle), find the differential coefficient of the following functions.

Answer / How to write in exam

Method (first principle): $f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$. Below is the method shown fully for one representative function from each family; the rest follow exactly the same technique and are standard results to memorise.

Worked example – $f(x) = \sin x$:

$$f'(x) = \lim_{h \rightarrow 0} \frac{\sin(x+h) - \sin x}{h} = \lim_{h \rightarrow 0} \frac{2 \cos\left(x + \frac{h}{2}\right) \sin \frac{h}{2}}{h} = \lim_{h \rightarrow 0} \cos\left(x + \frac{h}{2}\right) \cdot \frac{\sin(h/2)}{h/2} = \cos x \cdot 1 = \cos x$$

Worked example – $f(x) = e^x$:

$$f'(x) = \lim_{h \rightarrow 0} \frac{e^{x+h} - e^x}{h} = e^x \lim_{h \rightarrow 0} \frac{e^h - 1}{h} = e^x \cdot 1 = e^x$$

(using the standard limit $\lim_{h \rightarrow 0} \frac{e^h - 1}{h} = 1$)

Worked example – $f(x) = \ln x$:

$$f'(x) = \lim_{h \rightarrow 0} \frac{\ln(x+h) - \ln x}{h} = \lim_{h \rightarrow 0} \frac{1}{h} \ln\left(1 + \frac{h}{x}\right) = \lim_{h \rightarrow 0} \frac{1}{x} \cdot \frac{\ln(1 + h/x)}{h/x} = \frac{1}{x} \cdot 1 = \frac{1}{x}$$

i. $\frac{d}{dx}(x^n) = nx^{n-1}$ $\frac{d}{dx}(a^x) = a^x \ln a$ $\frac{d}{dx}(\ln x) = \frac{1}{x}$ $\frac{d}{dx}(\log_a x) = \frac{1}{x \ln a}$

ii. $\frac{d}{dx}(e^x) = e^x$ $\frac{d}{dx}(e^{mx}) = me^{mx}$ $\frac{d}{dx}(e^{\sqrt{x}}) = e^{\sqrt{x}} \cdot \frac{1}{2\sqrt{x}}$

iii. (all by the same "compound angle" technique as $\sin x$ above)

$$\frac{d}{dx}(\sin x) = \cos x, \quad \frac{d}{dx}(\cos x) = -\sin x, \quad \frac{d}{dx}(\tan x) = \sec^2 x$$

$$\frac{d}{dx}(\cot x) = -\csc^2 x, \quad \frac{d}{dx}(\sec x) = \sec x \tan x, \quad \frac{d}{dx}(\csc x) = -\csc x \cot x$$

iv. (standard results for inverse trig)

$$\frac{d}{dx}(\sin^{-1} x) = \frac{1}{\sqrt{1-x^2}}, \quad \frac{d}{dx}(\cos^{-1} x) = \frac{-1}{\sqrt{1-x^2}}, \quad \frac{d}{dx}(\tan^{-1} x) = \frac{1}{1+x^2}$$

$$\frac{d}{dx}(\cot^{-1} x) = \frac{-1}{1+x^2}, \quad \frac{d}{dx}(\sec^{-1} x) = \frac{1}{|x|\sqrt{x^2-1}}, \quad \frac{d}{dx}(\csc^{-1} x) = \frac{-1}{|x|\sqrt{x^2-1}}$$

Q4. Differentiate the following functions (use standard rules: chain, product, quotient, logarithmic differentiation).

Answer / How to write in exam

i. $y = \sec(\tan^{-1} x)$. *Idea: draw a right triangle.* Let $\theta = \tan^{-1} x$, so $\tan \theta = x$, hypotenuse $= \sqrt{1+x^2}$, hence $\sec \theta = \sqrt{1+x^2}$.

$$y = \sqrt{1+x^2} \Rightarrow \frac{dy}{dx} = \boxed{\frac{x}{\sqrt{1+x^2}}}$$

ii. $y = \log [\sec(ax+b)]^3 = 3 \log \sec(ax+b)$.

$$\frac{dy}{dx} = 3 \cdot \frac{1}{\sec(ax+b)} \cdot \sec(ax+b) \tan(ax+b) \cdot a = \boxed{3a \tan(ax+b)}$$

iii. $y = \ln(x^2+2)$.

$$\frac{dy}{dx} = \boxed{\frac{2x}{x^2+2}}$$

iv. $y = 5x^{3/5} + \tan^2(2x)$.

$$\frac{dy}{dx} = 5 \cdot \frac{3}{5} x^{-2/5} + 2 \tan(2x) \sec^2(2x) \cdot 2 = \boxed{3x^{-2/5} + 4 \tan(2x) \sec^2(2x)}$$

v. $y = \log_a x + \log_x a$. Write $\log_a x = \frac{\ln x}{\ln a}$, $\log_x a = \frac{\ln a}{\ln x}$.

$$\frac{dy}{dx} = \frac{1}{x \ln a} - \frac{\ln a}{x(\ln x)^2}$$

$$\boxed{\frac{dy}{dx} = \frac{1}{x \ln a} - \frac{\ln a}{x(\ln x)^2}}$$

vi. $y = (\ln \sin x)^2$.

$$\frac{dy}{dx} = 2 \ln(\sin x) \cdot \frac{\cos x}{\sin x} = \boxed{2 \cot x \ln(\sin x)}$$

vii. $y = \frac{x^n}{\log_a x}$. Quotient rule with $\log_a x = \ln x / \ln a$:

$$\frac{dy}{dx} = \frac{nx^{n-1} \log_a x - x^n \cdot \frac{1}{x \ln a}}{(\log_a x)^2} = \boxed{\frac{x^{n-1} [n \log_a x - \frac{1}{\ln a}]}{(\log_a x)^2}}$$

viii. $y = \tan^{-1} \left(\frac{a+bx}{b-ax} \right)$. *Idea: recognise the tangent addition formula.* Write $\frac{a+bx}{b-ax} = \frac{\frac{a}{b} + x}{1 - \frac{a}{b}x}$,

which has the form $\frac{\tan \alpha + \tan \beta}{1 - \tan \alpha \tan \beta} = \tan(\alpha + \beta)$ with $\tan \alpha = \frac{a}{b}$, $\tan \beta = x$.

$$y = \tan^{-1} \left(\frac{a}{b} \right) + \tan^{-1} x$$

Since $\tan^{-1}(a/b)$ is a constant,

$$\frac{dy}{dx} = \boxed{\frac{1}{1+x^2}}$$

ix. $y = (x^2 + \sqrt{x})^{1/3}$.

$$\frac{dy}{dx} = \frac{1}{3} (x^2 + \sqrt{x})^{-2/3} \left(2x + \frac{1}{2\sqrt{x}} \right)$$

$$\boxed{\frac{dy}{dx} = \frac{1}{3} (x^2 + \sqrt{x})^{-2/3} \left(2x + \frac{1}{2\sqrt{x}} \right)}$$

x. $y = 10^{\ln \sin x}$.

$$\frac{dy}{dx} = 10^{\ln \sin x} \cdot \ln 10 \cdot \frac{\cos x}{\sin x} = \boxed{10^{\ln \sin x} \ln 10 \cot x}$$

xi. $y = e^{\sin x} - (\sin x^2)^2$.

$$\frac{dy}{dx} = e^{\sin x} \cos x - 2 \sin(x^2) \cos(x^2) \cdot 2x = \boxed{e^{\sin x} \cos x - 2x \sin(2x^2)}$$

xii. *These all need logarithmic differentiation: take $\ln$ of both sides first, then differentiate implicitly.*

• $y = x^{x^x}$: $\ln y = x^x \ln x$. Using $\frac{d}{dx}(x^x) = x^x(\ln x + 1)$:

$$\frac{y'}{y} = x^x(\ln x + 1) \ln x + \frac{x^x}{x} \Rightarrow y' = x^{x^x} \cdot x^x \left[(\ln x + 1) \ln x + \frac{1}{x} \right]$$

• $y = x^{e^x}$: $\ln y = e^x \ln x \Rightarrow \frac{y'}{y} = e^x \ln x + \frac{e^x}{x} \Rightarrow y' = x^{e^x} e^x \left(\ln x + \frac{1}{x} \right)$

• $y = (\sin x)^{\ln x}$: $\ln y = \ln x \ln(\sin x) \Rightarrow \frac{y'}{y} = \frac{\ln(\sin x)}{x} + \ln x \cot x \Rightarrow y' = (\sin x)^{\ln x} \left[\frac{\ln(\sin x)}{x} + \ln x \cot x \right]$

• $y = x^{\cos^{-1} x}$: $\ln y = \cos^{-1} x \ln x \Rightarrow \frac{y'}{y} = \frac{-\ln x}{\sqrt{1-x^2}} + \frac{\cos^{-1} x}{x} \Rightarrow y' = x^{\cos^{-1} x} \left[\frac{\cos^{-1} x}{x} - \frac{\ln x}{\sqrt{1-x^2}} \right]$

• $y = (\sin^{-1} x)^{\ln x}$: $\ln y = \ln x \ln(\sin^{-1} x) \Rightarrow \frac{y'}{y} = \frac{\ln(\sin^{-1} x)}{x} + \frac{\ln x}{(\sin^{-1} x)\sqrt{1-x^2}}$

$$\Rightarrow y' = (\sin^{-1} x)^{\ln x} \left[\frac{\ln(\sin^{-1} x)}{x} + \frac{\ln x}{(\sin^{-1} x)\sqrt{1-x^2}} \right]$$

xiii. $y = \sin^2(\ln x^2)$. Since $\frac{d}{dx} \ln x^2 = \frac{2}{x}$:

$$\frac{dy}{dx} = 2 \sin(\ln x^2) \cos(\ln x^2) \cdot \frac{2}{x} = \boxed{\frac{2}{x} \sin(2 \ln x^2)}$$

xiv. $y = (\sec \tan^{-1} x) (x^5 + x^{1/3})^{5/3}$. From part (i), $\sec(\tan^{-1} x) = \sqrt{1+x^2}$. Let $u = \sqrt{1+x^2}$, $v = (x^5 + x^{1/3})^{5/3}$ and use the product rule $y' = u'v + uv'$:

$$u' = \frac{x}{\sqrt{1+x^2}}, \quad v' = \frac{5}{3} (x^5 + x^{1/3})^{2/3} \left(5x^4 + \frac{1}{3}x^{-2/3} \right)$$

$$\boxed{\frac{dy}{dx} = \frac{x(x^5 + x^{1/3})^{5/3}}{\sqrt{1+x^2}} + \frac{5}{3}\sqrt{1+x^2}(x^5 + x^{1/3})^{2/3}\left(5x^4 + \frac{1}{3}x^{-2/3}\right)}$$

xv. $y = \sin\left(2 \tan^{-1} \sqrt{\frac{1-x}{1+x}}\right)$. *Idea: substitute $x = \cos \theta$. Then $\sqrt{\frac{1-\cos \theta}{1+\cos \theta}} = \tan \frac{\theta}{2}$, so $\tan^{-1} \sqrt{\frac{1-x}{1+x}} = \frac{\theta}{2}$, hence $2 \tan^{-1}(\dots) = \theta = \cos^{-1} x$.*

$$y = \sin(\cos^{-1} x) = \sqrt{1-x^2} \Rightarrow \frac{dy}{dx} = \boxed{\frac{-x}{\sqrt{1-x^2}}}$$

xvi. $y = \sin^{-1}\left(\frac{x + \sqrt{1-x^2}}{\sqrt{2}}\right)$. *Idea: substitute $x = \sin \varphi$. Then $\sqrt{1-x^2} = \cos \varphi$, and*

$$\frac{\sin \varphi + \cos \varphi}{\sqrt{2}} = \sin\left(\varphi + \frac{\pi}{4}\right)$$

$$y = \sin^{-1}\left[\sin\left(\varphi + \frac{\pi}{4}\right)\right] = \varphi + \frac{\pi}{4} = \sin^{-1} x + \frac{\pi}{4}$$

$$\frac{dy}{dx} = \boxed{\frac{1}{\sqrt{1-x^2}}}$$

xvii. $\sqrt{\frac{x}{y}} + \sqrt{\frac{y}{x}} = 1$. Multiply throughout by $\sqrt{xy}$: $x + y = \sqrt{xy}$. Differentiate implicitly w.r.t. x :

$$1 + y' = \frac{1}{2\sqrt{xy}}(y + xy')$$

Multiply by $2\sqrt{xy}$ and use $\sqrt{xy} = x + y$:

$$2(x + y) + 2(x + y)y' = y + xy'$$

$$2(x + y) - y = xy' - 2(x + y)y'$$

$$2x + y = y'[x - 2(x + y)] = y'(-x - 2y)$$

$$\boxed{\frac{dy}{dx} = -\frac{2x + y}{x + 2y}}$$

Q5. If $y = \sqrt{\cos x + \sqrt{\cos x + \sqrt{\cos x + \dots \infty}}}$, find $\frac{dy}{dx}$.

Answer / How to write in exam

Idea: the infinite nested radical repeats itself, so the whole expression under the first root equals y again:

$$y = \sqrt{\cos x + y} \Rightarrow y^2 = \cos x + y$$

Differentiate implicitly w.r.t. x :

$$2yy' = -\sin x + y' \Rightarrow y'(2y - 1) = -\sin x$$

$$\boxed{\frac{dy}{dx} = \frac{-\sin x}{2y - 1}}$$

Q6. If $y = \sqrt{\sin x + \sqrt{\sin x + \sqrt{\sin x + \cdots \infty}}}$, show that $y_1 = \frac{\cos x}{2y - 1}$.

Answer / How to write in exam

As above: $y^2 = \sin x + y$. Differentiate:

$$2yy_1 = \cos x + y_1 \Rightarrow y_1(2y - 1) = \cos x \Rightarrow y_1 = \frac{\cos x}{2y - 1} \quad \blacksquare$$

Q7. If $y = \cos x^{\cos x^{\cos x^{\cdots \infty}}}$, find $\frac{dy}{dx}$.

Answer / How to write in exam

Idea: the infinite tower of exponents equals y itself, so $y = (\cos x)^y$. Take $\ln$:

$$\ln y = y \ln(\cos x)$$

Differentiate implicitly:

$$\frac{y'}{y} = y' \ln(\cos x) + y(-\tan x)$$

$$y' \left[\frac{1}{y} - \ln(\cos x) \right] = -y \tan x$$

$$y' \cdot \frac{1 - y \ln(\cos x)}{y} = -y \tan x$$

$$\boxed{\frac{dy}{dx} = \frac{-y^2 \tan x}{1 - y \ln(\cos x)}}$$

Q8. Differentiate $\tan^{-1} \frac{\sqrt{1+x^2}-1}{x}$ with respect to $\sec^{-1} \frac{1}{2x^2-1}$.

Answer / How to write in exam

Idea: substitute $x = \tan \theta$ in both expressions to simplify them.

Let $u = \tan^{-1} \frac{\sqrt{1+x^2}-1}{x}$. With $x = \tan \theta$, $\sqrt{1+x^2} = \sec \theta$:

$$u = \tan^{-1} \left(\frac{\sec \theta - 1}{\tan \theta} \right) = \tan^{-1} \left(\tan \frac{\theta}{2} \right) = \frac{\theta}{2} = \frac{1}{2} \tan^{-1} x$$

$$\frac{du}{dx} = \frac{1}{2(1+x^2)}$$

Let $v = \sec^{-1} \frac{1}{2x^2 - 1} = \cos^{-1}(2x^2 - 1)$ (since $\sec^{-1}(1/y) = \cos^{-1} y$). Using $\cos^{-1}(2x^2 - 1) = 2 \cos^{-1} x$:

$$v = 2 \cos^{-1} x \Rightarrow \frac{dv}{dx} = \frac{-2}{\sqrt{1-x^2}}$$

$$\frac{du}{dv} = \frac{du/dx}{dv/dx} = \frac{\frac{1}{2(1+x^2)}}{\frac{-2}{\sqrt{1-x^2}}} = \boxed{\frac{-\sqrt{1-x^2}}{4(1+x^2)}}$$

Q9. Differentiate $\tan^{-1} \frac{2x}{1-x^2}$ **with respect to** $\sin^{-1} \frac{2x}{1+x^2}$.

Answer / How to write in exam

Substitute $x = \tan \theta$. Then $\frac{2x}{1-x^2} = \tan 2\theta$ and $\frac{2x}{1+x^2} = \sin 2\theta$.

$$u = \tan^{-1}(\tan 2\theta) = 2\theta = 2 \tan^{-1} x \Rightarrow \frac{du}{dx} = \frac{2}{1+x^2}$$

$$v = \sin^{-1}(\sin 2\theta) = 2\theta = 2 \tan^{-1} x \Rightarrow \frac{dv}{dx} = \frac{2}{1+x^2}$$

$$\frac{du}{dv} = \boxed{1}$$

Q10. If $C = 1 + r \cos \theta + \frac{r^2 \cos 2\theta}{2!} + \frac{r^3 \cos 3\theta}{3!} + \dots$ **and** $S = r \sin \theta + \frac{r^2 \sin 2\theta}{2!} + \frac{r^3 \sin 3\theta}{3!} + \dots$, **prove that** $C \frac{dC}{dr} + S \frac{dS}{dr} = (C^2 + S^2) \cos \theta$.

Answer / How to write in exam

Idea: recognise $C + iS$ as the exponential series.

$$C + iS = 1 + re^{i\theta} + \frac{(re^{i\theta})^2}{2!} + \frac{(re^{i\theta})^3}{3!} + \dots = e^{re^{i\theta}}$$

Differentiating w.r.t. r :

$$\frac{d}{dr}(C + iS) = e^{i\theta} \cdot e^{re^{i\theta}} = e^{i\theta}(C + iS) = (\cos \theta + i \sin \theta)(C + iS)$$

$$\text{So } \frac{dC}{dr} + i \frac{dS}{dr} = (C \cos \theta - S \sin \theta) + i(C \sin \theta + S \cos \theta).$$

Comparing real and imaginary parts:

$$\frac{dC}{dr} = C \cos \theta - S \sin \theta, \quad \frac{dS}{dr} = C \sin \theta + S \cos \theta$$

Now compute:

$$\begin{aligned} C \frac{dC}{dr} + S \frac{dS}{dr} &= C(C \cos \theta - S \sin \theta) + S(C \sin \theta + S \cos \theta) \\ &= C^2 \cos \theta - CS \sin \theta + SC \sin \theta + S^2 \cos \theta = (C^2 + S^2) \cos \theta \quad \blacksquare \end{aligned}$$

7 Successive Differentiation

Q1. Find the n th derivative of the following functions.

Answer / How to write in exam

Standard results (memorise these – write the $n = 1, 2, 3$ pattern in exam if asked to "derive"):

$$y = \ln x \Rightarrow y_n = (-1)^{n-1} \frac{(n-1)!}{x^n}$$

$$y = \frac{1}{x} \Rightarrow y_n = (-1)^n \frac{n!}{x^{n+1}}$$

$$y = a^x \Rightarrow y_n = a^x (\ln a)^n$$

$$y = e^{mx} \Rightarrow y_n = m^n e^{mx}$$

$$y = x^m \Rightarrow y_n = m(m-1) \cdots (m-n+1) x^{m-n}$$

$$y = \sin x \Rightarrow y_n = \sin \left(x + \frac{n\pi}{2} \right)$$

$$y = \cos x \Rightarrow y_n = \cos \left(x + \frac{n\pi}{2} \right)$$

$$y = \ln(ax+b) \Rightarrow y_n = (-1)^{n-1} (n-1)! \frac{a^n}{(ax+b)^n}$$

$$y = \sin(ax+b) \Rightarrow y_n = a^n \sin \left(ax+b + \frac{n\pi}{2} \right)$$

$$y = \cos(ax+b) \Rightarrow y_n = a^n \cos \left(ax+b + \frac{n\pi}{2} \right)$$

Q2. State and prove Leibnitz's theorem.

Answer / How to write in exam

Statement: if u, v are n -times differentiable functions of x , then

$$(uv)_n = \sum_{r=0}^n \binom{n}{r} u_{n-r} v_r = u_n v + \binom{n}{1} u_{n-1} v_1 + \binom{n}{2} u_{n-2} v_2 + \cdots + u v_n$$

Proof (by induction on n): Base case ($n = 1$): $(uv)_1 = u_1 v + u v_1 = \binom{1}{0} u_1 v_0 + \binom{1}{1} u_0 v_1$ — true (ordinary product rule).

Inductive step: assume true for $n = k$: $(uv)_k = \sum_{r=0}^k \binom{k}{r} u_{k-r} v_r$. Differentiate once more:

$$(uv)_{k+1} = \sum_{r=0}^k \binom{k}{r} (u_{k-r+1} v_r + u_{k-r} v_{r+1})$$

Collecting the coefficient of $u_{k+1-r} v_r$ from both parts and using Pascal's identity $\binom{k}{r} + \binom{k}{r-1} = \binom{k+1}{r}$, this simplifies to

$$(uv)_{k+1} = \sum_{r=0}^{k+1} \binom{k+1}{r} u_{k+1-r} v_r$$

which is the statement for $n = k + 1$. By induction, the theorem holds for all n . ■

Q3. If $y = x^n \log x$, prove that $y_{n+1} = \frac{n!}{x}$.

Answer / How to write in exam

Idea: apply Leibnitz's theorem to differentiate $u \cdot v = (\ln x)(x^n)$ a total of $n + 1$ times, taking $u = \ln x$ (differentiate many times), $v = x^n$ (vanishes after n derivatives).

$$u^{(k)} = \frac{(-1)^{k-1}(k-1)!}{x^k} \quad (k \geq 1), \quad v^{(k)} = \frac{n!}{(n-k)!} x^{n-k} \quad (k \leq n), \quad v^{(k)} = 0 \quad (k > n)$$

By Leibnitz's theorem,

$$y_{n+1} = (uv)^{(n+1)} = \sum_{k=1}^{n+1} \binom{n+1}{k} u^{(k)} v^{(n+1-k)}$$

(the $k = 0$ term vanishes since $v^{(n+1)} = 0$). Substituting and simplifying each term:

$$\binom{n+1}{k} u^{(k)} v^{(n+1-k)} = (-1)^{k-1} \frac{n! (n+1)!}{k! (n+1-k)!} \cdot \frac{1}{x}$$

Summing over $k = 1$ to $n + 1$ and using the binomial identity $\sum_{k=0}^{n+1} (-1)^k \binom{n+1}{k} = 0 \Rightarrow$

$$\sum_{k=1}^{n+1} (-1)^{k-1} \binom{n+1}{k} = 1:$$

$$y_{n+1} = \frac{n!}{x} \cdot 1 = \boxed{\frac{n!}{x}} \quad \blacksquare$$

Q4. If $y = \tan^{-1} x$, show that $(1 + x^2)y_{n+1} + 2nxy_n + n(n-1)y_{n-1} = 0$.

Answer / How to write in exam

$y' = \frac{1}{1+x^2} \Rightarrow (1+x^2)y' = 1$. Differentiate this n times using Leibnitz's theorem on the product $(1+x^2) \cdot y'$ (note $\frac{d^2}{dx^2}(1+x^2) = 2$, all higher derivatives of $1+x^2$ vanish, and the RHS "1" is constant so all its derivatives are 0):

$$(1+x^2)y_{n+1} + n(2x)y_n + \binom{n}{2}(2)y_{n-1} = 0$$

$$(1+x^2)y_{n+1} + 2nxy_n + n(n-1)y_{n-1} = 0 \quad \blacksquare$$

Q5. If $x = \cos \sqrt{y}$, show that $(1-x^2)y_{n+2} - (2n+1)xy_{n+1} - n^2y_n = 0$.

Answer / How to write in exam

$$x = \cos \sqrt{y} \Rightarrow \sqrt{y} = \cos^{-1} x \Rightarrow y = (\cos^{-1} x)^2.$$

$$y' = 2 \cos^{-1} x \cdot \left(\frac{-1}{\sqrt{1-x^2}} \right) \Rightarrow \sqrt{1-x^2} y' = -2 \cos^{-1} x$$

Square: $(1-x^2)y'^2 = 4(\cos^{-1} x)^2 = 4y$. Differentiate w.r.t. x :

$$-2xy'^2 + (1-x^2)2y'y'' = 4y'$$

Divide throughout by $2y'$:

$$-xy' + (1-x^2)y'' = 2 \Rightarrow (1-x^2)y'' - xy' - 2 = 0$$

Differentiate n times by Leibnitz's theorem (RHS constant so derivatives are 0):

$$(1-x^2)y_{n+2} - 2nxy_{n+1} - n(n-1)y_n - [xy_{n+1} + ny_n] = 0$$

$$(1-x^2)y_{n+2} - (2n+1)xy_{n+1} - n^2y_n = 0 \quad \blacksquare$$

Q6. If $y = e^{a \cos^{-1} x}$ (or $y = e^{a \sin^{-1} x}$), show that $(1-x^2)y_{n+2} - (2n+1)xy_{n+1} - (n^2+a^2)y_n = 0$.

Answer / How to write in exam

$$y' = \frac{-a}{\sqrt{1-x^2}} y \Rightarrow \sqrt{1-x^2} y' = -ay. \text{ Square: } (1-x^2)y'^2 = a^2y^2. \text{ Differentiate:}$$

$$-2xy'^2 + (1-x^2)2y'y'' = 2a^2yy'$$

$$\text{Divide by } 2y': \quad (1-x^2)y'' - xy' - a^2y = 0.$$

Differentiate n times (Leibnitz):

$$(1-x^2)y_{n+2} - 2nxy_{n+1} - n(n-1)y_n - xy_{n+1} - ny_n - a^2y_n = 0$$

$$(1-x^2)y_{n+2} - (2n+1)xy_{n+1} - (n^2+a^2)y_n = 0 \quad \blacksquare$$

Q7. If $x = \sin \left(\frac{1}{m} \ln y \right)$, show that $(1-x^2)y_{n+2} - (2n+1)xy_{n+1} - (n^2+m^2)y_n = 0$.

Answer / How to write in exam

$$x = \sin \left(\frac{1}{m} \ln y \right) \Rightarrow \sin^{-1} x = \frac{1}{m} \ln y \Rightarrow y = e^{m \sin^{-1} x}.$$

This is identical in form to Q6 with $a = m$: $y' = \frac{m}{\sqrt{1-x^2}} y$, leading (by the same steps) to $(1-x^2)y'' - xy' - m^2y = 0$, and after differentiating n times:

$$(1-x^2)y_{n+2} - (2n+1)xy_{n+1} - (n^2+m^2)y_n = 0 \quad \blacksquare$$

Q8. If $\ln y = \tan^{-1} x$, show that $(1 + x^2)y_{n+2} + (2nx + 2x - 1)y_{n+1} + n(n + 1)y_n = 0$.

Answer / How to write in exam

$\frac{y'}{y} = \frac{1}{1+x^2} \Rightarrow (1+x^2)y' = y$. Differentiate once:

$$(1+x^2)y'' + 2xy' = y' \Rightarrow (1+x^2)y'' + (2x-1)y' = 0$$

Differentiate n times (Leibnitz):

$$(1+x^2)y_{n+2} + 2nxy_{n+1} + n(n-1)y_n + (2x-1)y_{n+1} + 2ny_n = 0$$

$$(1+x^2)y_{n+2} + (2nx+2x-1)y_{n+1} + [n(n-1)+2n]y_n = 0$$

Since $n(n-1)+2n = n^2+n = n(n+1)$:

$$(1+x^2)y_{n+2} + (2nx+2x-1)y_{n+1} + n(n+1)y_n = 0 \quad \blacksquare$$

Q9. If $y = a \sin^{-1} x + b \cos^{-1} x$, show that $(1-x^2)y_{n+2} - (2n+1)xy_{n+1} - n^2y_n = 0$.

Answer / How to write in exam

$y' = \frac{a}{\sqrt{1-x^2}} - \frac{b}{\sqrt{1-x^2}} = \frac{a-b}{\sqrt{1-x^2}}$, a constant times $(1-x^2)^{-1/2}$.

$\sqrt{1-x^2}y' = a-b$ (constant). Square: $(1-x^2)y'^2 = (a-b)^2$. Differentiate:

$$-2xy'^2 + (1-x^2)2y'y'' = 0 \Rightarrow (1-x^2)y'' - xy' = 0$$

Differentiate n times:

$$(1-x^2)y_{n+2} - 2nxy_{n+1} - n(n-1)y_n - xy_{n+1} - ny_n = 0$$

$$(1-x^2)y_{n+2} - (2n+1)xy_{n+1} - n^2y_n = 0 \quad \blacksquare$$

Q10. If $y = f(x) = \sin^{-1} x$, show that

(i) $(1-x^2)y_2 - xy_1 = 0$ (ii) $(1-x^2)y_{n+2} - (2n+1)xy_{n+1} - n^2y_n = 0$ (iii) $(y_n)_0 = (n-2)^2(n-4)^2 \dots 3^2 \cdot 1^2$ for odd n (iv) $(y_n)_0 = 0$ for even n .

Answer / How to write in exam

(i) $y' = \frac{1}{\sqrt{1-x^2}} \Rightarrow \sqrt{1-x^2}y' = 1$. Square: $(1-x^2)y'^2 = 1$. Differentiate:

$$-2xy'^2 + (1-x^2)2y'y'' = 0$$

Divide by $2y'$: $(1-x^2)y'' - xy' = 0$, i.e. $(1-x^2)y_2 - xy_1 = 0$. $\blacksquare$

(ii) Differentiate the result of (i) n times, exactly as in Q9:

$$(1-x^2)y_{n+2} - (2n+1)xy_{n+1} - n^2y_n = 0 \quad \blacksquare$$

(iii), (iv) Put $x = 0$ in the recurrence of (ii): $y_{n+2}(0) = n^2 y_n(0)$.

Since $\sin^{-1} x$ is an **odd function**, its Maclaurin series has only odd powers of x , so **every even-order derivative at $x = 0$ is zero**: $(y_n)_0 = 0$ for even n . This proves (iv).

For odd n , start from $y(0) = 0$, $y_1(0) = 1$ and apply $y_{n+2}(0) = n^2 y_n(0)$ repeatedly:

$$y_3(0) = 1^2 \cdot y_1(0) = 1^2, \quad y_5(0) = 3^2 \cdot y_3(0) = 3^2 \cdot 1^2, \quad y_7(0) = 5^2 \cdot 3^2 \cdot 1^2, \dots$$

In general, for odd n :

$$(y_n)_0 = (n-2)^2(n-4)^2 \dots 3^2 \cdot 1^2 \quad \blacksquare$$

8 General Theorem & Expansion

8.1 Rolle's Theorem

Q1. State and prove Rolle's theorem.

Answer / How to write in exam

Statement: If a function $f(x)$ is

1. continuous on the closed interval $[a, b]$,
2. differentiable on the open interval (a, b) , and
3. $f(a) = f(b)$,

then there exists at least one point $c \in (a, b)$ such that $f'(c) = 0$.

Proof: Since f is continuous on $[a, b]$, by the Extreme Value Theorem it attains a maximum value M and a minimum value m on $[a, b]$.

Case 1: If $M = m$, then f is constant on $[a, b]$, so $f'(x) = 0$ for every $x \in (a, b)$ – done.

Case 2: If $M \neq m$, since $f(a) = f(b)$, at least one of M, m must be attained at some interior point $c \in (a, b)$ (it cannot be that both extremes occur only at the equal endpoints). At this interior extreme point, since f is differentiable, by Fermat's theorem $f'(c) = 0$. ■

Q2 & Q5. Verify Rolle's theorem for $f(x) = 2x^3 + x^2 - 4x - 2$ over $[-\sqrt{2}, \sqrt{2}]$.

Answer / How to write in exam

Idea: factorise first — $f(x) = x^2(2x + 1) - 2(2x + 1) = (2x + 1)(x^2 - 2)$, so the roots are $x = -\frac{1}{2}, \pm\sqrt{2}$.

Continuity & differentiability: f is a polynomial, so continuous on $[-\sqrt{2}, \sqrt{2}]$ and differentiable on $(-\sqrt{2}, \sqrt{2})$. ✓

Equal end values:

$$f(-\sqrt{2}) = (2(-\sqrt{2}) + 1)(2 - 2) = 0, \quad f(\sqrt{2}) = (2\sqrt{2} + 1)(2 - 2) = 0$$

$f(-\sqrt{2}) = f(\sqrt{2}) = 0$. ✓ So Rolle's theorem applies.

Find c : $f'(x) = 6x^2 + 2x - 4 = 0 \Rightarrow 3x^2 + x - 2 = 0 \Rightarrow (3x - 2)(x + 1) = 0 \Rightarrow x = \frac{2}{3}, -1$

Both $c = \frac{2}{3}$ and $c = -1$ lie in $(-\sqrt{2}, \sqrt{2}) \approx (-1.414, 1.414)$.

Rolle's theorem is verified; $c = \frac{2}{3}$ or $c = -1$

Q3. Verify Rolle's theorem for $f(x) = x^2 - 3x + 2$ in the interval $(1, 2)$.

Answer / How to write in exam

$f(x) = x^2 - 3x + 2$ is a polynomial (continuous, differentiable everywhere).

$$f(1) = 1 - 3 + 2 = 0, \quad f(2) = 4 - 6 + 2 = 0$$

$$f(1) = f(2). \quad \checkmark$$

$$f'(x) = 2x - 3 = 0 \Rightarrow x = \frac{3}{2}, \text{ which lies in } (1, 2).$$

Verified; $c = \frac{3}{2}$

Q4. Determine whether Rolle's theorem can be applied to $f(x) = x^4 - 2x^2$ on $[-2, 2]$, and if so find all c .

Answer / How to write in exam

Polynomial $\Rightarrow$ continuous on $[-2, 2]$, differentiable on $(-2, 2)$.

$$f(-2) = 16 - 8 = 8, \quad f(2) = 16 - 8 = 8$$

$$f(-2) = f(2). \quad \checkmark \text{ So Rolle's theorem **can be applied**.}$$

$$f'(x) = 4x^3 - 4x = 4x(x^2 - 1) = 0 \Rightarrow x = 0, 1, -1, \text{ all lying in } (-2, 2).$$

$c = -1, 0, 1$

Q6. Verify Rolle's theorem for $f(x) = e^x \sin x$ in the interval $(0, \pi)$.

Answer / How to write in exam

f is a product of continuous, differentiable functions $\Rightarrow$ continuous on $[0, \pi]$, differentiable on $(0, \pi)$.

$$f(0) = e^0 \sin 0 = 0, \quad f(\pi) = e^\pi \sin \pi = 0$$

$$f(0) = f(\pi) = 0. \quad \checkmark$$

$$f'(x) = e^x \sin x + e^x \cos x = e^x (\sin x + \cos x) = 0 \Rightarrow \tan x = -1 \Rightarrow x = \frac{3\pi}{4} \in (0, \pi)$$

Verified; $c = \frac{3\pi}{4}$

8.2 Mean Value Theorem (Lagrange's)

Q1. State and prove the Mean Value Theorem.

Answer / How to write in exam

Statement: If $f(x)$ is continuous on $[a, b]$ and differentiable on (a, b) , then there exists $c \in (a, b)$ such that

$$f'(c) = \frac{f(b) - f(a)}{b - a}$$

Proof: Define an auxiliary function

$$g(x) = f(x) - f(a) - \frac{f(b) - f(a)}{b - a}(x - a)$$

Then $g(a) = 0$ and $g(b) = f(b) - f(a) - [f(b) - f(a)] = 0$, so $g(a) = g(b)$. Also g is continuous on $[a, b]$ and differentiable on (a, b) (being built from f and a linear function). By **Rolle's theorem**, there exists $c \in (a, b)$ with $g'(c) = 0$:

$$g'(x) = f'(x) - \frac{f(b) - f(a)}{b - a} \Rightarrow f'(c) = \frac{f(b) - f(a)}{b - a} \quad \blacksquare$$

Q2. Verify MVT for $f(x) = 3 + 2x - x^2$ in the interval $(0, 1)$.

Answer / How to write in exam

$$f(0) = 3, \quad f(1) = 3 + 2 - 1 = 4. \quad f'(x) = 2 - 2x.$$

$$f'(c) = \frac{f(1) - f(0)}{1 - 0} = \frac{4 - 3}{1} = 1 \Rightarrow 2 - 2c = 1 \Rightarrow c = \frac{1}{2}$$

$$c = \frac{1}{2} \in (0, 1). \quad \boxed{\text{Verified; } c = \frac{1}{2}}$$

Q3. Verify MVT for $f(x) = 3x^3 + 7x^2 - 11x - 15$.

Answer / How to write in exam

(The interval was not legible in the source image; the roots of f are $x = -3, -1, \frac{5}{3}$, since $f(x) = (x + 1)(3x^2 + 4x - 15) = (x + 1)(3x - 5)(x + 3)$. Using the natural interval $[-1, \frac{5}{3}]$ between two roots, where $f = 0$ at both ends, illustrates the method – use the same steps with your book's actual interval.)

$$f(-1) = 0, \quad f\left(\frac{5}{3}\right) = 0. \quad f'(x) = 9x^2 + 14x - 11.$$

$$f'(c) = \frac{0 - 0}{\frac{5}{3} - (-1)} = 0 \Rightarrow 9c^2 + 14c - 11 = 0 \Rightarrow c = \frac{-14 \pm \sqrt{196 + 396}}{18} = \frac{-14 \pm \sqrt{592}}{18}$$

Taking the root inside the interval: $c \approx 0.574 \in (-1, \frac{5}{3})$.

$$\boxed{\text{Verified (method shown); } c \approx 0.574}$$

Q4. Verify MVT for $f(x) = \frac{x^3}{4} + 1$ over the interval $[0, 2]$.

Answer / How to write in exam

$$f(0) = 1, \quad f(2) = \frac{8}{4} + 1 = 3. \quad f'(x) = \frac{3x^2}{4}.$$

$$f'(c) = \frac{3-1}{2-0} = 1 \Rightarrow \frac{3c^2}{4} = 1 \Rightarrow c^2 = \frac{4}{3} \Rightarrow c = \frac{2}{\sqrt{3}}$$

$$c = \frac{2}{\sqrt{3}} \approx 1.1547 \in (0, 2). \quad \boxed{\text{Verified; } c = \frac{2}{\sqrt{3}}}$$

Q5. Verify MVT for $f(x) = 2x^3 - 8x + 1$ when $a = 1$, $b = 3$.

Answer / How to write in exam

$$f(1) = 2 - 8 + 1 = -5, \quad f(3) = 54 - 24 + 1 = 31. \quad f'(x) = 6x^2 - 8.$$

$$f'(c) = \frac{31 - (-5)}{3 - 1} = \frac{36}{2} = 18 \Rightarrow 6c^2 - 8 = 18 \Rightarrow c^2 = \frac{13}{3} \Rightarrow c = \sqrt{\frac{13}{3}}$$

$$c \approx 2.08 \in (1, 3). \quad \boxed{\text{Verified; } c = \sqrt{13/3}}$$

Q6. Verify MVT for $f(x) = x^2 - 5x + 7$ in $1 \leq x \leq 3$.

Answer / How to write in exam

$$f(1) = 1 - 5 + 7 = 3, \quad f(3) = 9 - 15 + 7 = 1. \quad f'(x) = 2x - 5.$$

$$f'(c) = \frac{1-3}{3-1} = -1 \Rightarrow 2c - 5 = -1 \Rightarrow c = 2$$

$$c = 2 \in (1, 3). \quad \boxed{\text{Verified; } c = 2}$$

Q7. Verify MVT for $f(x) = x(x-1)(x-2)$ in the interval $(0, 0.5)$.

Answer / How to write in exam

$$\text{Expand: } f(x) = x^3 - 3x^2 + 2x.$$

$$f(0) = 0, \quad f(0.5) = 0.125 - 0.75 + 1 = 0.375$$

$$f'(x) = 3x^2 - 6x + 2.$$

$$f'(c) = \frac{0.375 - 0}{0.5} = 0.75 \Rightarrow 3c^2 - 6c + 2 = 0.75 \Rightarrow 12c^2 - 24c + 5 = 0$$

$$c = \frac{24 \pm \sqrt{576 - 240}}{24} = \frac{24 \pm \sqrt{336}}{24}$$

Taking the value in $(0, 0.5)$: $c \approx 0.236$.

Verified; $c \approx 0.236$

8.3 Taylor's Series

Q1. State Taylor's series.

Answer / How to write in exam

If $f(x)$ has derivatives of all orders at $x = a$, then

$$f(x) = f(a) + f'(a)(x-a) + \frac{f''(a)}{2!}(x-a)^2 + \frac{f'''(a)}{3!}(x-a)^3 + \cdots = \sum_{n=0}^{\infty} \frac{f^{(n)}(a)}{n!}(x-a)^n$$

Q2. Expand $\frac{1}{x}$ in ascending powers of $(x-2)$ by Taylor's series.

Answer / How to write in exam

$f(x) = \frac{1}{x}$, $a = 2$. Compute derivatives:

$$f(x) = x^{-1}, \quad f'(x) = -x^{-2}, \quad f''(x) = 2x^{-3}, \quad f'''(x) = -6x^{-4}, \quad \dots, \quad f^{(n)}(x) = (-1)^n \frac{n!}{x^{n+1}}$$

At $x = 2$: $f^{(n)}(2) = (-1)^n \frac{n!}{2^{n+1}}$.

Taylor coefficient: $\frac{f^{(n)}(2)}{n!}(x-2)^n = \frac{(-1)^n}{2^{n+1}}(x-2)^n$.

$$\frac{1}{x} = \frac{1}{2} - \frac{(x-2)}{4} + \frac{(x-2)^2}{8} - \frac{(x-2)^3}{16} + \cdots = \sum_{n=0}^{\infty} \frac{(-1)^n (x-2)^n}{2^{n+1}}$$

Q3. Expand $\ln x$ in ascending powers of $(x-2)$ by Taylor's series.

Answer / How to write in exam

$f(x) = \ln x$, $a = 2$, $f(2) = \ln 2$.

$$f'(x) = \frac{1}{x} \Rightarrow f'(2) = \frac{1}{2}, \quad f''(x) = -\frac{1}{x^2} \Rightarrow f''(2) = -\frac{1}{4}, \quad f'''(x) = \frac{2}{x^3} \Rightarrow f'''(2) = \frac{1}{4}$$

In general $f^{(n)}(2) = (-1)^{n-1} \frac{(n-1)!}{2^n}$ for $n \geq 1$, so the coefficient $\frac{f^{(n)}(2)}{n!} = \frac{(-1)^{n-1}}{n \cdot 2^n}$.

$$\ln x = \ln 2 + \frac{(x-2)}{2} - \frac{(x-2)^2}{8} + \frac{(x-2)^3}{24} - \cdots = \ln 2 + \sum_{n=1}^{\infty} \frac{(-1)^{n-1} (x-2)^n}{n \cdot 2^n}$$

8.4 Maclaurin's Series

Q8. State and prove Maclaurin's theorem.**Answer / How to write in exam**

Statement: Maclaurin's series is Taylor's series about $a = 0$:

$$f(x) = f(0) + f'(0)x + \frac{f''(0)}{2!}x^2 + \frac{f'''(0)}{3!}x^3 + \cdots = \sum_{n=0}^{\infty} \frac{f^{(n)}(0)}{n!}x^n$$

Proof: Set $a = 0$ directly in Taylor's theorem (proved by repeatedly applying the Mean Value Theorem / Rolle's theorem to the remainder term, or by successive integration by parts of $f(x) - f(0) = \int_0^x f'(t) dt$). Substituting $a = 0$ into the general Taylor expansion

$$f(x) = f(a) + f'(a)(x-a) + \frac{f''(a)}{2!}(x-a)^2 + \cdots$$

immediately gives the Maclaurin form above. ■

Q9. Obtain the Maclaurin series generated by $f(x) = \log(1+x)$, e^x , e^{mx} , $\sin x$, $\cos x$.**Answer / How to write in exam**

$f(x) = \ln(1+x)$: $f(0) = 0$; $f'(x) = \frac{1}{1+x} \Rightarrow f'(0) = 1$; $f''(x) = \frac{-1}{(1+x)^2} \Rightarrow f''(0) = -1$; $f'''(0) = 2$; in general $f^{(n)}(0) = (-1)^{n-1}(n-1)!$.

$$\ln(1+x) = x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + \cdots = \sum_{n=1}^{\infty} \frac{(-1)^{n-1}x^n}{n}$$

$f(x) = e^x$: all derivatives equal e^x , so $f^{(n)}(0) = 1$ for every n .

$$e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \cdots = \sum_{n=0}^{\infty} \frac{x^n}{n!}$$

$f(x) = e^{mx}$: $f^{(n)}(x) = m^n e^{mx} \Rightarrow f^{(n)}(0) = m^n$.

$$e^{mx} = 1 + mx + \frac{(mx)^2}{2!} + \frac{(mx)^3}{3!} + \cdots = \sum_{n=0}^{\infty} \frac{(mx)^n}{n!}$$

$f(x) = \sin x$: $f(0) = 0$, $f'(0) = \cos 0 = 1$, $f''(0) = -\sin 0 = 0$, $f'''(0) = -\cos 0 = -1$, pattern repeats every 4 (only odd-order derivatives are nonzero).

$$\sin x = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + \cdots = \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n+1}}{(2n+1)!}$$

$f(x) = \cos x$: $f(0) = 1$, $f'(0) = 0$, $f''(0) = -1$, $f'''(0) = 0$ (only even-order derivatives nonzero).

$$\cos x = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!} + \cdots = \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n}}{(2n)!}$$

*Note: $\ln x$ has **no** Maclaurin series, because $\ln x$ (and all its derivatives) is undefined at $x = 0$ – Maclaurin expansion requires the function to be defined and differentiable at 0. (That is exactly why Q2/Q3 above expand $\ln x$ and $1/x$ about $x = 2$ instead, using Taylor's series.)*

Recommended books (from your sheet):

1. *A Text Book on Differential Calculus* – P. K. Bhattacharjee
2. *Differential Calculus* – P. N. Chatterjee